Microsoft Elevate

CAPSTONE PROJECT

Road Accident Analysis Using Power BI Dashboard

PRESENTED BY

STUDENT NAME:Navya Madagowni

COLLEGE NAME:Vardhaman College Of Engineering

DEPARTMENT:Computer Science And Engineering

EMAIL ID:navya250505@gmail.com

OUTLINE:

- **Problem Statement** (Should not include solution)
- **Proposed System/Solution**
- **System Development Approach** (Technology Used)
- **Algorithm & Deployment**
- **Result (Output Image)**
- **Conclusion**
- **Future Scope**
- **References**

PROBLEM STATEMENT:

- Example: Road accidents are a major concern worldwide, leading to injuries, fatalities, and property damage. Governments and traffic authorities require accurate insights to understand accident patterns and take preventive measures.
- However, raw accident datasets are difficult to analyze manually due to their large size and complexity. Without proper visualization and analytics, it becomes challenging to identify key factors such as accident locations, severity levels, road types, weather conditions, and time patterns.
- Therefore, there is a need for an interactive analytical system that can transform raw accident data into meaningful insights to help decision-makers reduce road accidents and improve traffic safety.

PROPOSED SOLUTION:

The proposed system uses **Power BI** to analyze road accident datasets and generate interactive dashboards for better decision-making.

- The system performs the following tasks:
- **Data Collection**
- Road accident data is collected from available datasets that contain details such as:
- Accident date
- Location
- Number of casualties
- Road type
- Weather conditions
- Vehicle type
- **Data Processing**
- The dataset is cleaned and transformed using **Power Query Editor** in Power BI.
- Steps include:
- Removing missing values
- Formatting date fields
- Creating calculated columns and measures

PROPOSED SOLUTION:

- **Data Visualization**
- Power BI dashboards are created to visualize accident trends including:
- Total casualties
- Fatal casualties
- Serious casualties
- Slight casualties
- Monthly accident trends
- Casualties by road type
- Casualties by light conditions
- Urban vs Rural accidents
- Geographic accident distribution using map visuals
- These visualizations help identify patterns and risk areas.

SYSTEM APPROACH:

- **Software Tools**
  - **Power BI Desktop**
  - **Microsoft Excel**
- **Technologies Used**
  - Data Visualization
  - Business Intelligence
  - Data Analytics
- **Power BI Features Used**
  - Power Query Editor
  - Data Modeling
  - DAX Measures
  - Interactive Dashboard
  - Map Visualization
- **Dataset**

  Road Accident Dataset containing accident records, location data, casualty details, and road conditions.

ALGORITHM & DEPLOYMENT:

The system follows a **data analytics workflow**:

- **Data Input**
  - Import accident dataset from Excel into Power BI.
- **Data Cleaning**
  - Remove duplicates
  - Handle missing values
  - Format data types.
- **Data Transformation**
  - Create calculated measures using DAX.

Example Measures:

- Total Casualties
- CY Casualties
- PY Casualties
- Fatal Casualties
- Serious Casualties
- Slight Casualties

ALGORITHM & DEPLOYMENT:

Visualization

- Create charts and KPIs to display insights.

Deployment

- The Power BI dashboard is used to present insights interactively.

Users can filter the dashboard by:

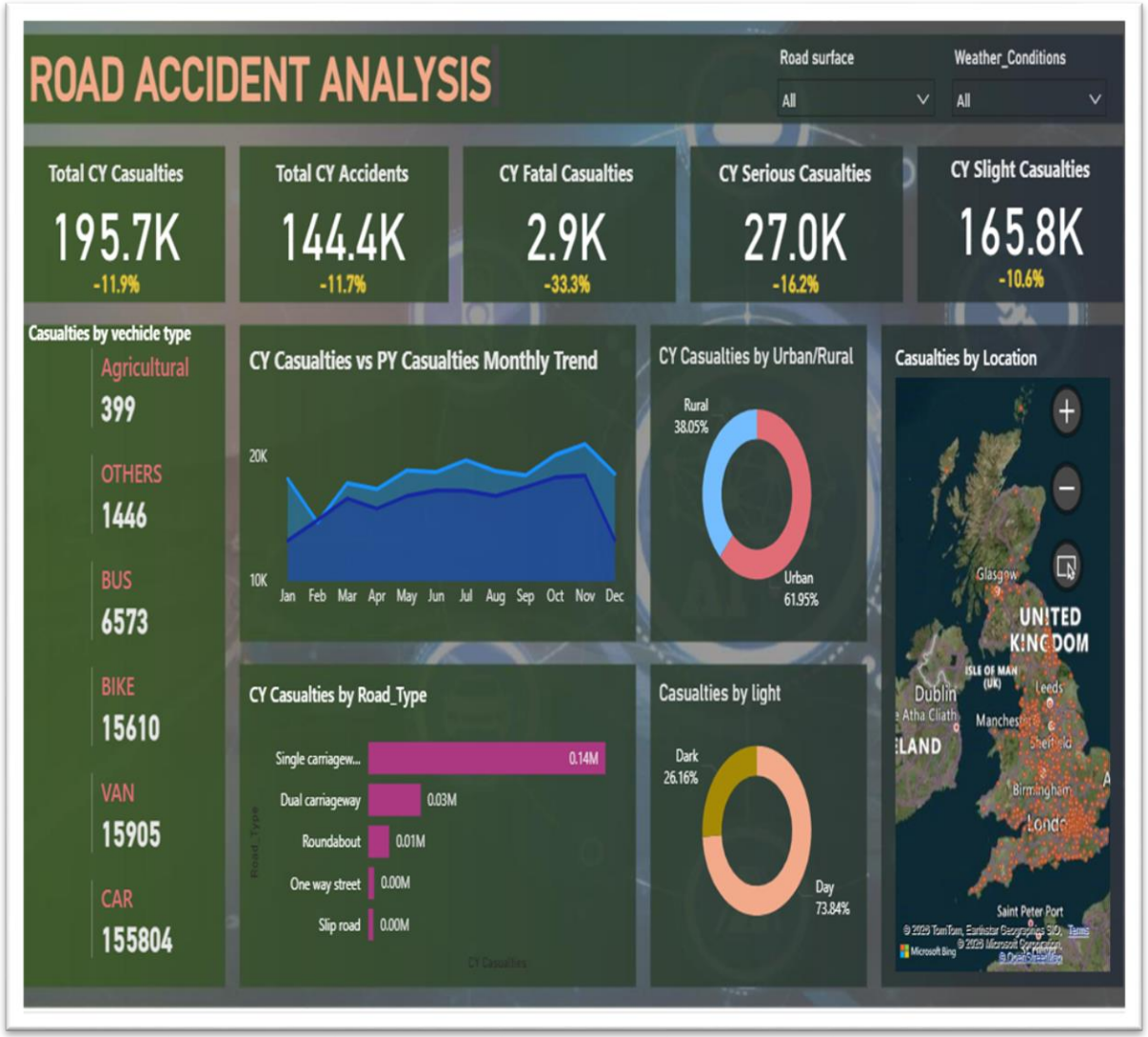
- Year
- Road type
- Location
- Weather condition

RESULT:

Insert screenshots of your Power BI dashboard:

- Examples:
- Road Accident Analysis Dashboard
- Casualties by Road Type Chart
- Monthly Accident Trend
- Urban vs Rural Chart
- Map showing accident locations

RESULT:


ROAD ACCIDENT ANALYSIS
Road surface
All
Weather_Conditions
All
Total CY Casualties
195.7K
-11.9%
Total CY Accidents
144.4K
-11.7%
CY Fatal Casualties
2.9K
-33.3%
CY Serious Casualties
27.0K
-16.2%
CY Slight Casualties
165.8K
-10.6%
Casualties by vechicle type
Agricultural
399
OTHERS
1446
BUS
6573
BIKE
15610
VAN
15905
CAR
155804
CY Casualties vs PY Casualties Monthly Trend
20K
10K
Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
CY Casualties by Urban/Rural
Rural
38.05%
Urban
61.95%
Casualties by Location
CY Casualties by Road_Type
Single carriagew...
0.14M
Dual carriageway
0.03M
Roundabout
0.01M
One way street
0.00M
Slip road
0.00M
Casualties by light
Dark
26.16%
Day
73.84%

ROAD ACCIDENT ANALYSIS
Road surface
All
Weather_Conditions
All
Total CY Casualties
15.9K
-9.5%
Total CY Accidents
11.7K
-10.8%
CY Fatal Casualties
0.2K
-38.7%
CY Serious Casualties
2198
-16.3%
CY Slight Casualties
13.5K
-7.4%
Agricultural
399
OTHERS
1446
BUS
6573
BIKE
15610
VAN
15905
CAR
155804
CY Casualties vs PY Casualties Monthly Trend
1500
1000
Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
CY Casualties by Urban/Rural
Rural
3.15%
Urban 4.98%
Casualties by Location
CY Casualties by Road_Type
Single carriagew...
0.01M
Dual carriageway
0.00M
Roundabout
0.00M
One way street
0.00M
Slip road
0.00M
Casualties by light
Dark
2.17%
Day 5.96%

RESULT:

ROAD ACCIDENT ANALYSIS
Road surface
All
Weather_Conditions
All
Total CY Casualties
6573
5.6%
Total CY Accidents
4932
7.8%
CY Fatal Casualties
0.1K
-57.6%
CY Serious Casualties
912
1.8%
CY Slight Casualties
5611
7.7%
Casualties by vehicle type
Agricultural
399
OTHERS
1446
BUS
6573
BIKE
15610
VAN
15905
CAR
155804
CY Casualties vs PY Casualties Monthly Trend
600
400
Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
CY Casualties by Urban/Rural
Rural
1.03%
Urban 2.32%
Casualties by Location
Glasgow
UNITED KINGDOM
ISLE OF MAN (UK)
Dublin
Leeds
Manchester
Sheffield
Birmingham
London
Saint Peter Port
CY Casualties by Road_Type
Single carriagew...
0.00M
Dual carriageway
0.00M
Roundabout
0.00M
One way street
0.00M
Slip road
0.00M
Casualties by light
Dark
0.89%
Day 2.47%

ROAD ACCIDENT ANALYSIS
Road surface
All
Weather_Conditions
All
Total CY Casualties
15.6K
-14.0%
Total CY Accidents
11.8K
-11.1%
CY Fatal Casualties
0.2K
-51.6%
CY Serious Casualties
2170
-18.0%
CY Slight Casualties
13.3K
-12.4%
Casualties by vehicle type
Agricultural
399
OTHERS
1446
BUS
6573
BIKE
15610
VAN
15905
CAR
155804
CY Casualties vs PY Casualties Monthly Trend
1500
1000
Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
CY Casualties by Urban/Rural
Rural
2.81%
Urban 5.16%
Casualties by Location
Glasgow
UNITED KINGDOM
ISLE OF MAN (UK)
Dublin
Leeds
Manchester
Sheffield
Birmingham
London
Saint Peter Port
CY Casualties by Road_Type
Single carriagew...
0.01M
Dual carriageway
0.00M
Roundabout
0.00M
One way street
0.00M
Slip road
0.00M
Casualties by light
Dark 2.1%
Day 5.87%

RESULT:

ROAD ACCIDENT ANALYSIS
Road surface
All
Weather_Conditions
All
Total CY Casualties
155.8K
-12.3%
Total CY Accidents
114.6K
-12.3%
CY Fatal Casualties
2.4K
-30.4%
CY Serious Casualties
21.5K
-16.5%
CY Slight Casualties
132.0K
-11.2%
Casualties by vehicle type
Agricultural
399
OTHERS
1446
BUS
6573
BIKE
15610
VAN
15905
CAR
155804
CY Casualties vs PY Casualties Monthly Trend
15K
10K
Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
CY Casualties by Urban/Rural
Rural
30.7%
Urban 48.9%
Casualties by Location
CY Casualties by Road_Type
Single carriagew...
0.12M
Dual carriageway
0.03M
Roundabout
0.01M
One way street
0.00M
Slip road
0.00M
Casualties by light
Dark
20.76%
Day
58.84%

ROAD ACCIDENT ANALYSIS
Road surface
All
Weather_Conditions
All
Total CY Casualties
399
-37.0%
Total CY Accidents
299
-33.6%
CY Fatal Casualties
0.0K
-22.2%
CY Serious Casualties
61
-39.6%
CY Slight Casualties
331
-36.7%
Casualties by vehicle type
Agricultural
399
OTHERS
1446
BUS
6573
BIKE
15610
VAN
15905
CAR
155804
CY Casualties vs PY Casualties Monthly Trend
80
60
40
20
Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
CY Casualties by Urban/Rural
Rural
0.09%
Urban 0.12%
Casualties by Location
CY Casualties by Road_Type
Single carriagew...
0.00M
Dual carriageway
0.00M
Roundabout
0.00M
One way street
0.00M
Slip road
0.00M
Casualties by light
Dark
0.05%
Day 0.15%

CONCLUSION:

- The Road Accident Analysis dashboard provides valuable insights into accident patterns and contributing factors.
- Using Power BI visualization tools, the system helps identify high-risk areas, accident trends, and casualty statistics. This information can support authorities in improving road safety measures, traffic management, and accident prevention strategies.
- The dashboard simplifies complex accident data into easy-to-understand visual insights for better decision-making.

FUTURE SCOPE:

- The system can be enhanced further by:
- Integrating **real-time traffic accident data**
- Using **machine learning models to predict accident risk**
- Expanding analysis to **multiple cities or countries**
- Developing **mobile dashboards for traffic authorities**
- Adding **weather and traffic density data for deeper analysis**

REFERENCES:

1.Microsoft Power BI Documentation

2.Road Accident Dataset-Kaggle
google Link:
https://drive.google.com/drive/folders/1N8pcPIDIEBYK6tC_ovCKjLeOSzk6kHSV?usp=sharing

3.Data Visualization Best Practices

4.Microsoft Learn –Power BI Tutorials

GitHub Link: https://github.com/madagowninavya-2505/road-accident-analysis-powerbi

Thank You